

Spiking, Recurrent, and Attention Autoencoders for Temporal Signals: A Leakage-Controlled Nested Cross-Validation Comparison

André Furlan

Instituto de Biociências, Letras e Ciências Exatas - UNESP
andre.furlan@unesp.br

Abstract—Spiking Neural Networks (SNNs) promise low-power temporal signal processing through event-driven, sparse computation, but their use as autoencoders for waveform reconstruction lacks a controlled, leakage-free empirical comparison against modern non-spiking baselines. We compare four trained autoencoder families — a spiking autoencoder (SNN-AE), an LSTM autoencoder (LSTM-AE), a GRU autoencoder (GRU-AE), and a bottlenecked Transformer autoencoder (Transformer-AE) without encoder-decoder cross-attention — under three spike-encoding front ends (direct, Poisson, latency), together with PCA and mean-frame linear reference points. Each family runs its own multi-objective (NSGA-II) architecture search; only the latent bottleneck width is held fixed across families, per dataset signal domain (16 for the two spoken-digit datasets, 64 for the two EEG datasets), so no family can win by evolving itself a wider bottleneck than the others. All models share a single C++/CPU harness [1] for matched timing and a transparent operation-count cost budget. Evaluation uses *nested six-fold leave-one-group-out cross-validation* on four datasets — spoken digits (FSDD [2] and AudioMNIST) and single-channel EEG (the PhysioNet EEG Motor Movement/Imagery Database and the Siena Scalp EEG Database) — where the group is the speaker or the EEG subject: no source recording or group crosses a split, and architecture and hyperparameters for every family are selected inside each fold on the held-out validation group only, then retrained on the combined training and validation data before a single evaluation on the untouched test group. Model differences are assessed per dataset with a recording-level paired analysis (bootstrap confidence intervals over recordings, Wilcoxon signed-rank, Holm-Bonferroni), with group-level and seed-level results reported as generalization- and optimization-robustness evidence. We report reconstruction quality (MSE, MAE, R^2) as mean $\pm$ standard deviation over seeds, a raw per-class operation count as the primary cost figure (with a weighted proxy as a labelled sensitivity analysis), the self-attention $O(T^2 d_{\text{model}})$ interaction cost, and real parameter counts. The harness and profiles are released for reproduction.

Index Terms—spiking neural networks, LSTM, GRU, Transformer, autoencoder, temporal signals, surrogate gradients, nested cross-validation, data leakage, audio signal processing, EEG, neuroevolution

I. INTRODUCTION

One-dimensional temporal signals such as audio waveforms, biosignals, and sensor streams often need to be processed and interpreted in low-power embedded systems where the energy budget is a primary constraint.

Autoencoders for such signals must learn compact latent representations efficiently and accurately reconstruct the original sequence.

Long Short-Term Memory (LSTM) autoencoders are a well-established baseline for sequence reconstruction [3], [4], but their dense matrix operations are computationally expensive. Spiking Neural Networks (SNNs) otherwise communicate via sparse binary events and perform synaptic updates only when spikes occur, offering potential energy reduction on general-purpose and neuromorphic hardware [5]–[7].

Training SNNs for reconstruction requires backpropagation through non-differentiable spike functions which behave like step functions. Surrogate gradient methods [8], [9] enable end-to-end gradient-based training and have produced competitive SNN classification performance [6], [10]. However, a controlled study of SNN autoencoders for temporal signal reconstruction — compared against several modern non-spiking baselines, under an evaluation protocol that provably prevents information from the test signals leaking into training or hyperparameter selection — is lacking.

Evaluation rigor is the crux. When fixed-length windows from all recordings are pooled, shuffled, and split, windows from the same utterance — and the same speaker — appear in both training and validation, so reported errors partly measure memorization of speaker- and recording-specific waveform detail. Speaker (or subject) identity is the natural unit of generalization, and the publicly available spoken-digit benchmarks used here have on the order of tens of such groups (six and sixty), which also bounds how strong any population-level claim can be; the two EEG benchmarks span a wider range (fourteen and one hundred nine subjects). We therefore evaluate on four datasets and make the recording-level paired difference, not the group-level one, the primary estimand.

A second source of unfairness is architectural: if one family is allowed a wider latent bottleneck than another, a reconstruction-quality difference may reflect only that difference in task difficulty, not architecture quality. We address this by giving every family its own architecture search rather than a shared or hand-picked shape, while holding the one dimension that sets the difficulty of the compression task — the latent width — fixed across all four families for a given dataset.

This paper addresses these gaps with four contributions:

- 1) A leakage-controlled protocol: *nested six-fold leave-one-group-out cross-validation* on four datasets (FSDD [2], AudioMNIST, the PhysioNet EEG Motor Movement/Imagery Database, the Siena Scalp EEG Database) in which no source recording or group (speaker / subject) crosses a split and architecture and hyperparameters are selected on the inner validation group only.
- 2) A comparison of four independently architecture-searched autoencoder families under three input encodings — SNN-AE, LSTM-AE, GRU-AE, and a bottlenecked Transformer-AE without encoder–decoder cross-attention — plus PCA and mean-frame linear reference points, all on one matched C++/CPU harness [1], [11]. Each family runs its own multi-objective (NSGA-II) search over its own architecture genes; only the latent width is fixed, per dataset signal domain, so that a difference in reconstruction quality cannot be explained by one family being handed an easier compression task.
- 3) A hierarchical statistical analysis with the recording-level paired difference as the primary estimand (bootstrap over recordings, Wilcoxon, Holm–Bonferroni), and group-level and seed-level results reported as robustness rather than population inference.
- 4) An analytic computational-cost budget reported as a raw per-class operation count (primary) with a weighted proxy as a labelled sensitivity analysis, alongside the self-attention $O(T^2 d_{\text{model}})$ term and real parameter counts; the reproducible framework is released.

II. RELATED WORK

A. Recurrent and attention sequence autoencoders

LSTM autoencoders encode variable-length sequences into fixed-size latent vectors and reconstruct the original sequence from the latent code [3], [4]. The constant-error carousel in the cell recurrence mitigates vanishing gradients during backpropagation through time (BPTT) [12], [13]. The Gated Recurrent Unit (GRU) [14] merges the input and forget gates and drops the separate cell state, giving a smaller recurrent cell with performance comparable to the LSTM on many sequence tasks [15]. Self-attention encoders [16] replace recurrence with a content-based $O(T^2)$ mixing operation and layer normalization [17]. We use all three as trained non-spiking baselines; the Transformer variant is *bottlenecked* — the decoder is conditioned only on the fixed-dimension latent, with no encoder–decoder cross-attention — so that, like PCA and the recurrent autoencoders, all reconstruction information must pass through the latent.

B. Spiking neural networks and surrogate gradients

SNNs model computation via binary spike events governed by membrane potential dynamics [5], [7]. The Leaky Integrate-and-Fire (LIF) neuron is the most widely deployed model. Training with backpropagation requires differentiable surrogates for the discontinuous Heaviside activation; surrogate gradient methods [8], [18], [19] have enabled competitive SNN accuracy with lower inference cost. SNN research has also expanded

from classification to temporal sequence modelling and representation learning, but systematic reconstruction benchmarks remain limited [10], [19].

C. Spike encoding for temporal signals

Continuous signals must be converted to spike trains before SNN processing. Rate coding (Poisson spike generation proportional to amplitude), latency coding (earlier spikes for stronger stimuli), and direct pass-through of normalized values represent the three dominant strategies [8], [20]. Encoding choice affects sparsity, gradient flow, training stability, and reconstruction fidelity [8].

D. Evaluation protocol and leakage

On speaker-labelled audio benchmarks, pooling windows before splitting places windows from the same utterance and speaker on both sides of the split, so measured error conflates generalization with memorization. Speaker-disjoint (leave-one-speaker-out) evaluation is the standard remedy; when model or hyperparameter selection is also involved, a *nested* scheme is required so that the test speaker never influences selection or early stopping [21]. Three of the four benchmarks here have tens of groups (six, sixty, fourteen); the fourth has one hundred nine, still a modest population by classification-dataset standards. We treat the group-level result as a robustness check throughout and make the recording-level paired difference the primary estimand.

III. METHODOLOGY

A. Problem formulation

Let $x \in \mathbb{R}^T$ denote a temporal signal window of length T . An autoencoder learns encoder $E : \mathbb{R}^T \rightarrow \mathbb{R}^d$ and decoder $D : \mathbb{R}^d \rightarrow \mathbb{R}^T$ minimizing reconstruction loss:

$$\mathcal{L} = \|x - D(E(x))\|_2^2. \quad (1)$$

Where x is the input window, $E(x)$ is the latent representation, and $D(E(x))$ is the reconstructed output.

The latent dimension d is fixed *per dataset signal domain* rather than globally: $d = 16$ for the two spoken-digit datasets and $d = 64$ for the two EEG datasets (a 16:1 and a 4:1 compression ratio against the $T = 256$ window, respectively). Within a dataset d is identical for all four trained families and is never itself searched, so a reconstruction-quality difference reflects architecture quality at a matched compression ratio rather than a difference in task difficulty. Every other architectural dimension — hidden width H (SNN-AE, LSTM-AE, GRU-AE), encoder depth, d_{model} , attention heads, and feed-forward width (Transformer-AE) — is instead a gene in that family’s own architecture search (Sec. III-E); a per-family lower bound on that gene enforces $H, d_{\text{model}} \geq d$, so no searched architecture’s true bottleneck can fall below the dataset’s latent width through an undersized hidden layer upstream of it (Sec. III-E). The window is reshaped into T/f frames of f samples before entering the sequence models. PCA and a mean-frame predictor provide linear reference points, fitted per fold on training data only.

B. LIF neuron model

The continuous-time Leaky Integrate-and-Fire neuron is described by [7]:

$$\tau \frac{dV}{dt} = -(V - V_{\text{rest}}) + RI(t), \quad \tau = RC. \quad (2)$$

Where V is the membrane potential, V_{rest} is the resting potential, $I(t)$ is the input current, R is the membrane resistance, and C is the membrane capacitance. The time constant τ governs the rate of potential decay. The $RI(t)$ term is consistent with the discrete update in Eq. (3).

Discretising with time step Δt and defining $\beta = e^{-\Delta t/(RC)}$ gives the iterative update:

$$V[t] = \beta V[t-1] + RI[t]. \quad (3)$$

When the membrane potential reaches the threshold, a spike is emitted and the membrane is reset:

$$s[t] = \mathbf{1}[V[t] \geq V_{th}], \quad V \leftarrow \begin{cases} 0 & \text{(hard reset)} \\ V - V_{th} & \text{(soft reset)} \end{cases}. \quad (4)$$

Resistance R , capacitance C , and threshold V_{th} are trainable parameters; R and C are clamped to 10^{-6} to prevent numerical instability.

Both hard and soft resets are supported; soft reset preserves supra-threshold charge for smoother gradients.

C. Surrogate gradient training

The spike function $s = \mathbf{1}[V \geq V_{th}]$ has true gradient zero almost everywhere, blocking standard backpropagation [8].

Surrogate gradient methods replace $\partial s / \partial V$ with a smooth approximation in the backward pass while preserving the exact Heaviside rule in the forward pass [8], [18].

Exponential (SuperSpike [9]):

$$\hat{\sigma}'(V) = \frac{1}{\sigma_s} \exp\left(-\frac{|V - V_{th}|}{\sigma_s}\right). \quad (5)$$

Exponential surrogate gradient. σ_s controls the width of the approximation.

Boxcar:

$$\hat{\sigma}'(V) = \mathbf{1}[|V - V_{th}| < \frac{w}{2}]. \quad (6)$$

Boxcar surrogate gradient. w controls the width of the approximation.

ArcTan [10]:

$$\hat{\sigma}'(V) = \frac{\alpha}{2 \left(1 + \left(\frac{\pi}{2} \alpha (V - V_{th})\right)^2\right)}. \quad (7)$$

ArcTan surrogate gradient, the default spike-derivative estimator in `snnTorch` since 2023. Compared to the exponential and boxcar surrogates above, its heavier tails keep gradient flowing further from threshold, avoiding the saturation those surrogates show away from V_{th} .

We use the arctan surrogate ($\alpha = 2$). BPTT [12] unrolls both LSTM and SNN through the full time dimension T ; the reverse-time recurrence for layer l is:

$$\delta_l[t] = \hat{\sigma}'(V[t]) \odot (W_{l+1}^\top \delta_{l+1}[t]) + \beta \delta_l[t+1], \quad (8)$$

where $\delta_l[t] \in \mathbb{R}^{n_l}$ is the error signal at layer l and time t , $W_{l+1} \in \mathbb{R}^{n_{l+1} \times n_l}$ is the feed-forward weight matrix mapping layer- l activations to layer- $(l+1)$ pre-activations (so the forward pass computes $W_{l+1} a_l$), $\odot$ is the element-wise product, $\hat{\sigma}'(V[t])$ is the surrogate derivative of Sec. III-C, and $\beta = \partial V_{\text{post}} / \partial V_{\text{pre}} = e^{-\Delta t/(RC)}$ carries the error one step back in time through the membrane recurrence. The spatial term $W_{l+1}^\top \delta_{l+1}[t]$ and the temporal term $\beta \delta_l[t+1]$ are the SNN analogue of the standard BPTT split.

D. Recurrent and attention baselines

LSTM-AE. Standard gate equations — input, forget, and output gates from a dense pre-activation, with cell-state accumulation $C_t = f_t \odot C_{t-1} + i_t \odot g_t$ [4], [13]. The encoder stacks L LSTM layers of hidden width H (L and H searched, Sec. III-E); the final hidden state is projected to the d latent (fixed per dataset, Sec. IV) by $\text{Linear}(H \rightarrow d) \rightarrow \tanh$. The decoder applies $\text{Linear}(d \rightarrow H)$, replicates the result over time, runs L stacked LSTM layers, and projects each step back to the frame. States reset between samples.

GRU-AE. Identical topology with the LSTM cell replaced by a GRU cell [14], [15]: reset gate r_t , update gate z_t , and candidate $n_t = \tanh(x_t W_n^\top + r_t \odot (h_{t-1} U_n^\top) + b_n)$, $h_t = (1 - z_t) \odot n_t + z_t \odot h_{t-1}$. Dropping the cell state removes one gate's worth of parameters per layer; L and H are searched independently of LSTM-AE's (Sec. III-E), and the real counts are reported in the per-dataset summary tables (Table II and following).

Transformer-AE (bottlenecked). The encoder embeds each frame to a searched width d_{model} , adds sinusoidal positional encodings, applies N post-norm self-attention blocks with n_h heads [16], [17] (d_{model} , n_h , N , and the feed-forward width searched, Sec. III-E), mean-pools over time, and projects to the d latent through a $\tanh$. The decoder expands the latent, replicates it over T/f positions, adds positional encodings, applies N self-attention blocks, and projects back to frames. There is *no encoder-decoder cross-attention*: this is a deliberate design choice so that reconstruction must pass through the fixed-dimension latent, making the model a strict-compression autoencoder directly comparable to PCA and the recurrent autoencoders. Its parameter count is not matched to the recurrent baselines; the actual value is reported in the results.

Sequence-length cost. Per input window, the recurrent baselines cost $O(TdH)$ operations while self-attention costs $O(T^2 d_{\text{model}})$ for the score and context products plus $O(Td_{\text{model}}^2)$ for the projections. At $T/f = 32$ frames this quadratic interaction term is modest, but it is an architecture-level cost that is independent of the reconstruction-error ranking and is reported explicitly in the operation-count budget (Sec. III-J).

E. Architecture search

Every trained family — SNN-AE, LSTM-AE, GRU-AE, Transformer-AE — searches its own architecture with an independent multi-objective genetic algorithm (NSGA-II [22]), minimizing validation MSE and an inference-cost proxy; no

gene is shared across families and no population is shared across families. Each search uses a population of 8 for 5 generations (48 evaluations), re-scores every member of the final rank-0 Pareto front over 3 seeds to mitigate winner’s-curse selection bias, then picks the individual with the lowest re-scored validation MSE (ties broken by inference cost). The input encoding (direct, Poisson, latency) is itself a gene for every family, not an outer sweep dimension: a family’s search jointly selects its architecture *and* which encoding suits it best. SNN-AE additionally searches its pre-processing mode (Sec. III-G) and its encoder depth and per-layer width (strictly decreasing, terminating at the dataset’s latent width d); LSTM-AE and GRU-AE search hidden width $H \in [8, 256]$ and depth $L \in [1, 3]$; Transformer-AE searches $d_{\text{model}} \in [16, 128]$, attention heads from $\{1, 2, 4, 8\}$ (repaired to divide d_{model}), depth $N \in [1, 4]$, and feed-forward width in $[32, 256]$. The latent width d itself is never a gene, for any family (Sec. IV).

Bottleneck-floor constraint. Because H and d_{model} are searched independently of d , an otherwise-legal draw could place a family’s true information bottleneck *below* d — both LSTM-AE/GRU-AE and Transformer-AE project their recurrent state or embedding down to the latent through a separate linear layer, so $H < d$ or $d_{\text{model}} < d$ would make H or d_{model} , not d , that individual’s real bottleneck, silently invalidating the equal-compression-ratio comparison for that individual. Each family’s lower bound is therefore floored at $\max(\text{profile bound}, d)$ before sampling; SNN-AE’s own encoder-width repair enforces the analogous constraint structurally (every hidden width strictly exceeds d by construction). This floor changes the *effective* search range per dataset (tighter for the EEG datasets’ $d = 64$ than for the spoken-digit datasets’ $d = 16$) but never the profile’s stated bounds.

F. Spike encodings

Three strategies convert raw window x into pre-processed $\tilde{x}$:

Direct. $\tilde{x} = x$. Normalized values are passed without binarization.

Poisson. Each element is independently binarized via Bernoulli sampling [8], normalized over the window’s full range so that sub-mean (post-z-score negative) samples keep a nonzero firing probability rather than being clipped to zero:

$$\tilde{x}_i \sim \text{Bernoulli}\left(\text{clip}\left(\frac{x_i - x_{\min}}{x_{\max} - x_{\min}}, 0, 1\right)\right). \quad (9)$$

Where $x_{\min}$ and $x_{\max}$ are the minimum and maximum values in the window after z-score normalization, respectively — the same range used by the latency encoding below, so the two encodings see identical signal magnitude information and differ only in how it is temporally coded.

Latency. A single spike fires per element, at a time inversely proportional to signal strength (time-to-first-spike coding) [20], [23]:

$$\tilde{x}_{t,i} = \mathbf{1}\left[t = \left\lfloor \left(1 - \frac{x_i - x_{\min}}{x_{\max} - x_{\min}}\right) (T - 1) \right\rfloor\right]. \quad (10)$$

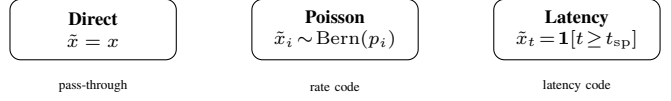


Figure 1. Three spike encoding schemes. Direct passes raw values; Poisson generates stochastic binary spikes proportional to amplitude; latency fires at a time inversely proportional to stimulus strength.

Where $x_{\min}$ and $x_{\max}$ are the minimum and maximum values in the window after z-score normalization, respectively. Stronger signals fire earlier; weaker signals fire later; each element fires exactly once.

G. SNN pre-processing modes

After spike encoding, a pre-processing transform ϕ is applied before the autoencoder input.

Dense. $\phi(\tilde{x}) = \tilde{x}$. No additional transform.

Conv1d. A fixed 3-tap temporal smoothing filter:

$$\phi(\tilde{x})_t = 0.25 \tilde{x}_{t-1} + 0.5 \tilde{x}_t + 0.25 \tilde{x}_{t+1}, \quad (11)$$

with boundary clamping. Reduces high-frequency spike jitter before autoencoder input.

Recurrent. A recurrent LIF cell integrates the encoded signal:

$$v[t] = \alpha v[t-1] + \tilde{x}[t] - s[t-1] V_{th}, \quad (12)$$

$$s[t] = \mathbf{1}[v[t] \geq V_{th}], \quad (13)$$

where α and V_{th} are sweep parameters.

H. SNN autoencoder architecture

The SNN autoencoder (Fig. 2) is a fully connected network with L encoder layers ($L \in [1, 4]$, searched, Sec. III-E) of strictly decreasing width, terminating at the dataset’s latent width d ; the decoder mirrors the encoder:

Enc: $\text{Lin}(T \rightarrow w_1) \rightarrow \text{Leaky} \rightarrow \dots \rightarrow \text{Lin}(w_{L-1} \rightarrow d) \rightarrow \text{Leaky}$

Dec: $\text{Lin}(d \rightarrow w_{L-1}) \rightarrow \dots \rightarrow \text{Lin}(w_1 \rightarrow T) \rightarrow \text{LeakyInt}$

with $T = 256$ and $w_1 > \dots > w_{L-1} > d$.

Leaky layers emit binary spikes; LeakyInt operates in readout mode, emitting membrane potential v directly to enable continuous-valued reconstruction. The discrete LIF update follows Eq. (3) with $R = 1/V_{th}$ and $C = -1/\ln(\alpha)$ derived from sweep parameters. All membrane states are reset between independent samples.

The four trained families are: SNN-AE (the spiking autoencoder above), LSTM-AE, GRU-AE, and Transformer-AE (Sec. III-D). The spiking pre-processing choices — dense (ϕ identity), conv1d (fixed 3-tap smoothing), recurrent (the LIF input transform above) — are *input transforms* to the one SNN autoencoder, not separate families: only the latent width d is fixed (per dataset); ϕ , the encoder depth and widths, the input encoding, and the biophysical parameters V_{th} and α are all jointly selected by the per-fold search procedure (Sec. III-E), so SNN-AE denotes the selected variant.

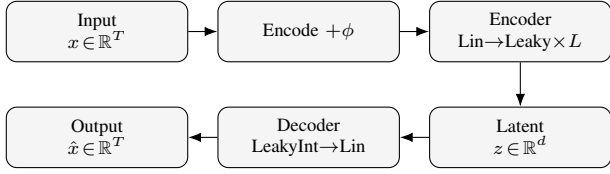


Figure 2. SNN autoencoder architecture. Input window x is encoded and pre-processed by mode ϕ , then compressed to latent $z \in \mathbb{R}^d$ (d fixed per dataset; L encoder layers, searched) by the encoder (Linear+Leaky layers). The decoder (LeakyInt+Linear) reconstructs $\hat{x}$.

I. Training

All trained models minimize MSE reconstruction loss via Adam [24] ($\eta = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$). SNN biophysical parameters (R , C , V_{th}) use a scaled learning rate $\eta_{bio} = 0.1\eta$ to stabilize membrane-dynamics optimization [8]; R and C are clamped to 10^{-6} in the forward pass with gradients zeroed in the clamped region. BPTT [12] propagates gradients through the full time dimension. Early stopping monitors a held-out monitor MSE with patience $P = 5$ epochs and minimum delta 10^{-8} over a maximum of 30 epochs [10]. Batch size is 1 to accommodate stateful sequence processing. Each configuration is trained with five random seeds (42–46).

Per-fold architecture selection (all four trained families). Within each outer fold, each of the four families runs its own architecture search (Sec. III-E) on that fold’s training groups, scoring every candidate on the inner validation group. Each family’s winner — the Pareto-front individual with the lowest 3-seed re-scored validation MSE, ties broken by inference cost — is then *retrained* on the training and validation groups combined and evaluated *once* on the held-out test group. Because train and validation groups share no recordings with the test group but not with each other, the retrain’s early-stopping monitor is a recording-disjoint slice carved (seeded, whole recordings) from the training groups, so no test-group signal ever reaches a fitting, search, or stopping decision. Each family’s per-fold winner and its inner-validation score are written to a selection manifest; the four winners are additionally compared against each other (lowest validation MSE, ties broken by inference cost) and the overall winner recorded per fold and repeat. SNN-AE’s selected pre-processing mode, V_{th} , and α are summarized per dataset in the selection tables (e.g. Table IV).

Linear references. PCA and the mean-frame predictor are not searched; both are fitted per fold on the training groups only, with no use of the validation or test group.

J. Analytic computational cost

We report an analytic *computational-cost* figure, not an energy measurement. The primary quantity is the raw operation count

$$C_{\text{raw}} = \sum_k N_k, \quad (14)$$

where N_k is the number of operations of class k executed in one forward pass over a window. We count, separately, the following classes: for the ANN models, dense multiply–accumulates

(MACs), nonlinearity evaluations, attention $QK^\top \cdot V$ products, and normalization operations; for the SNN, synaptic accumulations, neuron-state updates, and spike/event operations. The event-driven classes are scaled by the measured mean spike fraction $r \in [0, 1]$ of the evaluated run, so an SNN that never spikes contributes no event-operation cost.

As a secondary, explicitly labelled *sensitivity analysis* we also report a weighted proxy

$$C_{\text{weighted}} = \sum_k w_k N_k, \quad (15)$$

under a stated choice of relative per-class weights w_k ; the ranking of models under C_{weighted} is reported only alongside the weights that produce it, never as a standalone claim.

Neither C_{raw} nor C_{weighted} is energy. Real energy on a given device is dominated by memory movement, cache and SIMD behaviour, clock frequency, supply voltage, exploited sparsity, and the target silicon (CPU, GPU, or neuromorphic). On-device power measurement is future work; the figures here are design-time guidance for comparing architectures under matched software and a single CPU backend.

IV. EXPERIMENTAL SETUP

Datasets.

We evaluate on four datasets, each a corpus of 1-D signals with a natural group unit for leave-one-group-out (Table I): two spoken-digit corpora and two single-channel EEG corpora, chosen so the comparison spans two signal domains rather than variants of one task.

- **FSDD** [2] — spoken digits (0–9), six speakers, 8 kHz; group = speaker.
- **AudioMNIST** — spoken digits, sixty speakers, offline-resampled from 48 kHz to 8 kHz; group = speaker. The same task as FSDD at ten times the speaker count.
- **EEGMMIDB** [25], [26] — the PhysioNet EEG Motor Movement/Imagery Database, single-channel EEG (channel 0 of each recording), 109 subjects, 160 Hz; group = subject (one subject per source directory, so every recording of a subject stays in the same split by construction).
- **Siena** [26], [27] — the Siena Scalp EEG Database, single-channel EEG (channel 0), 14 adult epilepsy patients, 512 Hz; group = subject, same one-directory-per-subject grouping as EEGMMIDB.

Every recording is partitioned into non-overlapping windows of $T = 256$ samples with stride $S = T$ — 32 ms for the two audio datasets, 1.6 s for EEGMMIDB, and 0.5 s for Siena at their native sampling rates. For FSDD all windows are used; for the three larger datasets a per-recording cap keeps long recordings from dominating the pooled window counts (2 windows/recording for AudioMNIST, 40 for EEGMMIDB and Siena), which also keeps the four datasets’ training sizes comparable. Adjacent windows of one recording remain temporally correlated, which motivates the recording-level statistical unit (Sec. IV-A).

Preprocessing.

Table I

DATASETS. G = GROUP UNIT; K = OUTER FOLDS; CAP = WINDOWS KEPT PER RECORDING; d = LATENT WIDTH USED FOR THIS DATASET (SEC. IV).

Dataset	Signal	G	# G / K	cap	d
FSDD	digits, 8 kHz	speaker	6 / 6	—	16
AudioMNIST	digits, 8 kHz	speaker	60 / 6	2	16
EEGMMIDB	EEG, 160 Hz	subject	109 / 6	40	64
Siena	EEG, 512 Hz	subject	14 / 6	40	64

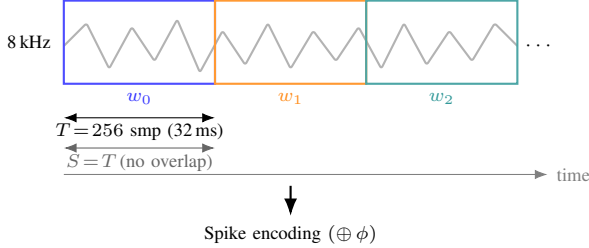


Figure 3. Signal windowing (audio case shown; $T = 256$ samples = 32 ms at 8 kHz). The EEG datasets use the same $T = 256$ at their own native rate (1.6 s at 160 Hz for EEGMMIDB, 0.5 s at 512 Hz for Siena). Stride $S = T$ yields windows with no shared samples. Windows are assigned to train / validation / test splits by group (speaker or EEG subject) before any pooling.

Each window is z-score normalised in place (subtract mean, divide by standard deviation) before encoding — a per-window statistic, so it introduces no cross-window leakage. Encodings that require a $[0, 1]$ range (Poisson, latency) re-normalise the z-scored window to its own $(x_{\min}, x_{\max})$ range internally. The three encodings are deterministic transforms (Poisson uses a per-window seeded draw) and are not fitted.

Nested leave-one-group-out cross-validation.

For each dataset the distinct groups are sorted and partitioned into $K = 6$ contiguous blocks. For outer fold $f \in \{0, \dots, 5\}$ the test block is f , the validation block is $(f + 1) \bmod 6$, and the remaining four blocks train. When $\#groups = K$ (FSDD) each block is a single speaker and this is ordinary leave-one-speaker-out; when $\#groups > K$ (AudioMNIST: 10 speakers per block; EEGMMIDB and Siena: ≈ 18 and ≈ 2 subjects per block, since 109 and 14 do not divide evenly by 6) it is grouped leave-one-block-out. Every window is assigned to a split *by its group, before any pooling*; training windows are shuffled (seeded), validation and test windows kept in (recording, position) order. Each group is in the test block exactly once and the validation block exactly once. At run start the harness asserts that no `recording_id` and no group appears in more than one split and writes a per-fold split manifest (group sets, per-recording window counts); a violation aborts the run. Headline numbers come only from the test split, aggregated across the six folds. Architecture and hyperparameters for every trained family are selected on the validation block only, then the winner is retrained on $\text{train} \cup \text{validation}$ before a single evaluation on the test block (Sec. III-E).

Evaluation harness.

The C++ harness [1] runs on the XTensor backend [11]

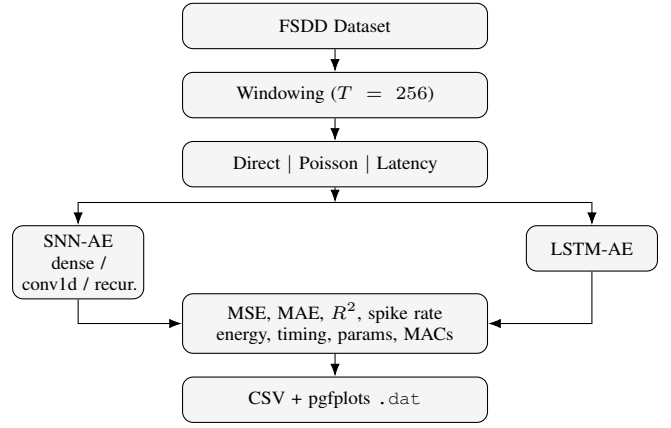


Figure 4. Experiment execution pipeline (FSDD shown; identical per dataset). Per outer fold, each window is encoded three ways and processed by the four independently architecture-searched trained families (each with its own per-fold NSGA-II selection) plus the PCA and mean-frame references. Per-window errors and manifests are written to CSV and JSON.

(CPU-only, CBLAS, SIMD via xsimd). All models — spiking and non-spiking — are trained and timed on this one backend, so the training and inference times in the per-dataset summary tables are directly comparable.

Metrics.

Per run we report reconstruction quality (MSE, MAE, R^2) and, for cost, spike rate (SNN only; 0 for the non-spiking models), the raw per-class operation count of Sec. III-J, parameter count, training time, and inference time in ms. FSDD provides no anomaly labels, so classification metrics are out of scope. Reconstruction error in the headline tables is the mean $\pm$ sample standard deviation over the five seeds (per-seed means first); the inferential analysis is in Sec. IV-A. A per-window error record (model, encoding, fold, split, group, recording, window, MSE, MAE) is written for every model so that paired comparisons can be formed at the window, recording, group, or seed level.

Profile.

A single JSON profile (`meeting01-los0.json`) configures the harness: the four datasets (each with its own latent width d , Table I), six outer folds, five seeds (42–46), the four trained families plus the two linear references, three encodings, and each family’s own NSGA-II architecture-search bounds (Sec. III-E). The run script iterates over (dataset, fold); a profile audit test verifies the profile parses and validates.

A. Statistical methods

Comparisons are paired on a shared, model-independent window identifier, so every model reconstructs the same windows and differences are formed per window. Let $\text{MSE}_{m,i}$ be the error of model m on window i .

Primary — recording level. For a recording r , let $E_{m,r} = \text{mean}_{i \in r} \text{MSE}_{m,i}$. For model m and reference b the paired difference is $d_r = E_{m,r} - E_{b,r}$. We report $\text{mean}(d_r)$, a 95% confidence interval from a bootstrap that resamples recordings, the Wilcoxon signed-rank p -value on

$\{d_r\}$, and Holm–Bonferroni correction across the reference set $b \in \{\text{LSTM-AE, GRU-AE, Transformer-AE, PCA, Mean}\}$. Effect size is reported both absolute ($\text{mean}(d_r)$) and relative ($\text{mean}(d_r) / \text{mean}(E_{b,\cdot})$). Recordings are the smallest unit across which windows are not, by construction, near-duplicates.

Robustness — group level. Per held-out group (speaker or EEG subject) we form the mean error and run a paired Wilcoxon / sign test. This checks whether the sign and rough magnitude of a difference hold across a dataset’s groups; with only tens to a little over a hundred groups it is not strong population-level inference and is reported as such.

Descriptive — window level. Paired bootstrap on all windows, reported but explicitly labelled pseudoreplicated (adjacent windows of a recording are correlated), so it is never the basis of a claim.

Robustness — seed level. Paired Wilcoxon across the five per-seed mean errors, characterizing optimization/reproducibility variation around the estimate.

The per-run CSVs are immutable; aggregation and statistics are separate downstream steps that never overwrite them. Synthetic known-answer tests (identical predictions $\Rightarrow p \approx 1$ and CI spanning 0; injected constant offset $\Rightarrow$ significant with CI excluding 0 and recovered offset; label permutation $\Rightarrow$ approximately uniform p) guard the analysis code in continuous integration.

V. RESULTS

Status. The tables and figure below are populated from the harness’s own output files by an automated pipeline (Sec. IV) and contain placeholder zeros until the full nested-LOSO grid has been run; the two EEG datasets additionally await download (Sec. VI). This draft therefore documents the finalized protocol and reporting structure ahead of that run, not yet its outcome.

All numbers come from the held-out test split only, pooled across the six outer folds, reported as mean $\pm$ sample standard deviation over the five seeds. Each dataset is reported in its own block: a per-model reconstruction and cost table, a per-encoding breakdown, the per-fold SNN selection, the recording-level paired analysis, and an MSE bar chart. The mean-frame predictor and PCA fix the scale of a “no-model” and a “best linear” reference; a trained model is interesting only to the extent it beats PCA, and an ordering among the trained families is asserted only where the Holm-corrected recording-level interval excludes zero. Because every trained family’s architecture is re-searched inside each fold (Sec. III-E), each model name denotes a *selection procedure*, not a fixed network; for SNN-AE, disagreement of the selected pre-processing mode across folds is itself a finding about how encoding-dependent the best pre-processing is. The cost columns (P , C_{raw} , timing) reflect each fold’s selected architecture and so can vary across folds and datasets, unlike in a fixed-architecture design; the Transformer’s $O(T^2 d_{\text{model}})$ self-attention term shows up in its C_{raw} independently of its MSE.

Table II

FSDD (SPOKEN DIGITS, 6 SPEAKERS): TEST RECONSTRUCTION AND COST BY MODEL (MEAN $\pm$ STD OVER FIVE SEEDS; BEST PER COLUMN IN BOLD). P = TRAINABLE PARAMETERS; C_{raw} = RAW PER-FORWARD-PASS OPERATION COUNT. PCA / MEAN ARE ANALYTIC REFERENCES.

Model	MSE	MAE	R^2	P	C_{raw}
Mean	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000			
PCA	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000			
SNN-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0
LSTM-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0
GRU-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0
Transformer-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0

Table III

FSDD (SPOKEN DIGITS, 6 SPEAKERS): RECONSTRUCTION PER INPUT ENCODING (MEAN $\pm$ STD OVER FIVE SEEDS).

Model	Enc.	MSE	MAE	R^2
SNN-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
LSTM-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
GRU-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
Transformer-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000

Table IV

FSDD (SPOKEN DIGITS, 6 SPEAKERS): SNN PRE-PROCESSING MODE / V_{th} / α SELECTED ON EACH FOLD’S VALIDATION BLOCK (PARENTHESES FRACTION = SEEDS NOT UNANIMOUS).

Fold	Test grp.	Val grp.	Enc.	Mode	V_{th}	α
0	—	—	—	—	—	—

Table V

FSDD (SPOKEN DIGITS, 6 SPEAKERS): LSTM-AE / GRU-AE ARCHITECTURE SELECTED BY NSGA-II ON EACH FOLD’S VALIDATION BLOCK – HIDDEN SIZE H , DEPTH L , AND INPUT ENCODING (MEDIAN / MODE OVER THE THREE RE-SCORING SEEDS; FAMILIES DISAMBIGUATED BY THEIR OWN INDEPENDENT SEARCH, NOT BY A SHARED COLUMN).

Family	Fold	Test spk.	Enc.	H	L	val MSE
—	0	—	—	—	—	—

Table VI

FSDD (SPOKEN DIGITS, 6 SPEAKERS): TRANSFORMER-AE ARCHITECTURE SELECTED BY NSGA-II ON EACH FOLD’S VALIDATION BLOCK – MODEL WIDTH d_{model} , HEADS, DEPTH N , FEED-FORWARD WIDTH d_{ff} , AND INPUT ENCODING (MEDIAN / MODE OVER THE THREE RE-SCORING SEEDS).

Fold	Test spk.	Enc.	d_{model}	Heads	N	d_{ff}
0	—	—	—	—	—	—

Table VII

FSDD (SPOKEN DIGITS, 6 SPEAKERS): RECORDING-LEVEL PAIRED ANALYSIS. $\bar{d}_r$ = MEAN PER-RECORDING MSE DIFFERENCE (MODEL – REFERENCE); NEGATIVE FAVOURS THE MODEL. CI = 95% BOOTSTRAP OVER RECORDINGS; p_{Holm} = HOLM-CORRECTED WILCOXON SIGNED-RANK.

Model	Reference	n_{rec}	$\bar{d}_r$	95% CI	p_{Holm}
—	—	0	0.0000	[0.0000, 0.0000]	1



Figure 5. FSDD (spoken digits, 6 speakers): test MSE by model and input encoding (mean over seeds and folds; mean-frame omitted for scale).

Table VIII

AUDIOMNIST (SPOKEN DIGITS, SIXTY SPEAKERS): TEST RECONSTRUCTION AND COST BY MODEL (MEAN $\pm$ STD OVER FIVE SEEDS; BEST PER COLUMN IN BOLD). P = TRAINABLE PARAMETERS; C_{raw} = RAW PER-FORWARD-PASS OPERATION COUNT. PCA / MEAN ARE ANALYTIC REFERENCES.

Model	MSE	MAE	R^2	P	$C_{\text{raw}}^{0.5}$
Mean	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000			
PCA	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000			
SNN-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0
LSTM-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0
GRU-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0
Transformer-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0

Table IX

AUDIOMNIST (SPOKEN DIGITS, SIXTY SPEAKERS): RECONSTRUCTION PER INPUT ENCODING (MEAN $\pm$ STD OVER FIVE SEEDS).

Model	Enc.	MSE	MAE	R^2
SNN-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
LSTM-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
GRU-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
Transformer-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000

Table X

AUDIOMNIST (SPOKEN DIGITS, SIXTY SPEAKERS): SNN PRE-PROCESSING MODE / V_{th} / α SELECTED ON EACH FOLD'S VALIDATION BLOCK (PARENTHESED FRACTION = SEEDS NOT UNANIMOUS).

Fold	Test grp.	Val grp.	Enc.	Mode	V_{th}	α
0	—	—	—	—	—	—

Table XI

AUDIOMNIST (SPOKEN DIGITS, SIXTY SPEAKERS): LSTM-AE / GRU-AE ARCHITECTURE SELECTED BY NSGA-II ON EACH FOLD'S VALIDATION BLOCK – HIDDEN SIZE H , DEPTH L , AND INPUT ENCODING (MEDIAN / MODE OVER THE THREE RE-SCORING SEEDS; FAMILIES DISAMBIGUATED BY THEIR OWN INDEPENDENT SEARCH, NOT BY A SHARED COLUMN).

Family	Fold	Test spk.	Enc.	H	L	val MSE
—	0	—	—	—	—	—

- FSDD (spoken digits, 6 speakers)
- AudioMNIST (spoken digits, sixty speakers)
- EEGMMIDB (EEG motor imagery, 109 subjects)
- Siena Scalp EEG (seizure recordings, 14 subjects)

Reading across datasets. A model family's advantage is credible to the extent it recurs on all four datasets' recording-

Table XII

AUDIOMNIST (SPOKEN DIGITS, SIXTY SPEAKERS): TRANSFORMER-AE ARCHITECTURE SELECTED BY NSGA-II ON EACH FOLD'S VALIDATION BLOCK – MODEL WIDTH d_{model} , HEADS, DEPTH N , FEED-FORWARD WIDTH d_{ff} , AND INPUT ENCODING (MEDIAN / MODE OVER THE THREE RE-SCORING SEEDS).

Fold	Test spk.	Enc.	d_{model}	Heads	N	d_{ff}
0	—	—	—	—	—	—

Table XIII

AUDIOMNIST (SPOKEN DIGITS, SIXTY SPEAKERS): RECORDING-LEVEL PAIRED ANALYSIS. $\bar{d}_r$ = MEAN PER-RECORDING MSE DIFFERENCE (MODEL – REFERENCE); NEGATIVE FAVOURS THE MODEL. CI = 95% BOOTSTRAP OVER RECORDINGS; p_{HOLM} = HOLM-CORRECTED WILCOXON SIGNED-RANK.

Model	Reference	n_{rec}	$\bar{d}_r$	95% CI	p_{Holm}
—	—	0	0.0000	[0.0000, 0.0000]	1

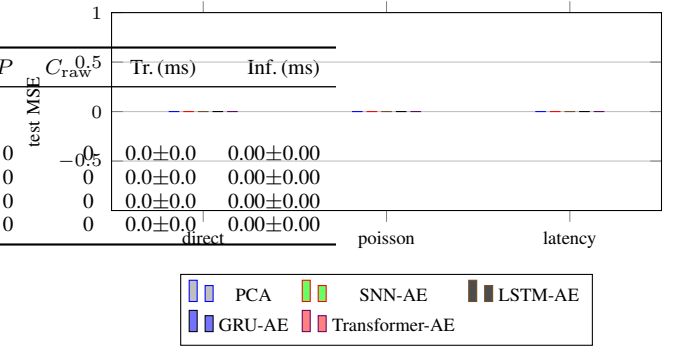


Figure 6. AudioMNIST (spoken digits, sixty speakers): test MSE by model and input encoding (mean over seeds and folds; mean-frame omitted for scale).

Table XIV

EEGMMIDB (EEG MOTOR IMAGERY, 109 SUBJECTS): TEST RECONSTRUCTION AND COST BY MODEL (MEAN $\pm$ STD OVER FIVE SEEDS; BEST PER COLUMN IN BOLD). P = TRAINABLE PARAMETERS; C_{raw} = RAW PER-FORWARD-PASS OPERATION COUNT. PCA / MEAN ARE ANALYTIC REFERENCES.

Model	MSE	MAE	R^2	P	C_{raw}
Mean	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000			
PCA	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000			
SNN-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0
LSTM-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0
GRU-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0
Transformer-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0

Table XV

EEGMMIDB (EEG MOTOR IMAGERY, 109 SUBJECTS): RECONSTRUCTION PER INPUT ENCODING (MEAN $\pm$ STD OVER FIVE SEEDS).

Model	Enc.	MSE	MAE	R^2
SNN-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
LSTM-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
GRU-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
Transformer-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000

Table XVI

EEGMMIDB (EEG MOTOR IMAGERY, 109 SUBJECTS): SNN PRE-PROCESSING MODE / V_{th} / α SELECTED ON EACH FOLD'S VALIDATION BLOCK (PARENTHESED FRACTION = SEEDS NOT UNANIMOUS).

Fold	Test grp.	Val grp.	Enc.	Mode	V_{th}	α
0	—	—	—	—	—	—

Table XVII

EEGMMIDB (EEG MOTOR IMAGERY, 109 SUBJECTS): LSTM-AE / GRU-AE ARCHITECTURE SELECTED BY NSGA-II ON EACH FOLD'S VALIDATION BLOCK – HIDDEN SIZE H , DEPTH L , AND INPUT ENCODING (MEDIAN / MODE OVER THE THREE RE-SCORING SEEDS; FAMILIES DISAMBIGUATED BY THEIR OWN INDEPENDENT SEARCH, NOT BY A SHARED COLUMN).

Family	Fold	Test spk.	Enc.	H	L	val MSE
—	0	—	—	—	—	—

Table XVIII

EEGMMIDB (EEG MOTOR IMAGERY, 109 SUBJECTS): TRANSFORMER-AE ARCHITECTURE SELECTED BY NSGA-II ON EACH FOLD'S VALIDATION BLOCK – MODEL WIDTH d_{model} , HEADS, DEPTH N , FEED-FORWARD WIDTH d_{ff} , AND INPUT ENCODING (MEDIAN / MODE OVER THE THREE RE-SCORING SEEDS).

Fold	Test spk.	Enc.	d_{model}	Heads	N	d_{ff}
0	—	—	—	—	—	—

Table XIX

EEGMMIDB (EEG MOTOR IMAGERY, 109 SUBJECTS): RECORDING-LEVEL PAIRED ANALYSIS. $\bar{d}_r$ = MEAN PER-RECORDING MSE DIFFERENCE (MODEL – REFERENCE); NEGATIVE FAVOURS THE MODEL. CI = 95% BOOTSTRAP OVER RECORDINGS; p_{HOLM} = HOLM-CORRECTED WILCOXON SIGNED-RANK.

Model	Reference	n_{rec}	$\bar{d}_r$	95% CI	p_{HOLM}
—	—	0	0.0000	[0.0000, 0.0000]	1

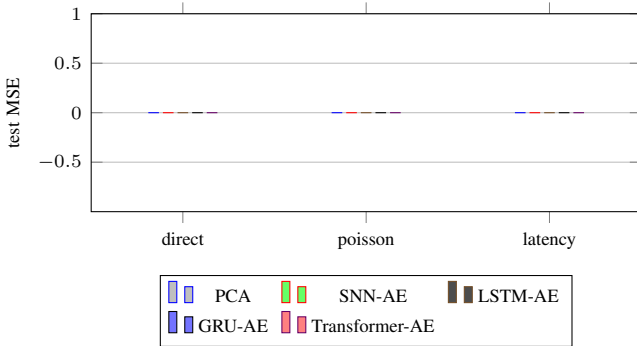


Figure 7. EEGMMIDB (EEG motor imagery, 109 subjects): test MSE by model and input encoding (mean over seeds and folds; mean-frame omitted for scale).

level analyses; an effect that appears only on FSDD — the smallest — or only on one modality is reported as such. The spoken-digit datasets (FSDD, AudioMNIST) test the same task at two speaker-pool sizes; EEGMMIDB and Siena test whether

Table XX

SIENA SCALP EEG (SEIZURE RECORDINGS, 14 SUBJECTS): TEST RECONSTRUCTION AND COST BY MODEL (MEAN $\pm$ STD OVER FIVE SEEDS; BEST PER COLUMN IN BOLD). P = TRAINABLE PARAMETERS; C_{raw} = RAW PER-FORWARD-PASS OPERATION COUNT. PCA / MEAN ARE ANALYTIC REFERENCES.

Model	MSE	MAE	R^2	P	C_{raw}
Mean	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000			
PCA	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000			
SNN-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0
LSTM-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0
GRU-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0
Transformer-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0

Table XXI

SIENA SCALP EEG (SEIZURE RECORDINGS, 14 SUBJECTS): RECONSTRUCTION PER INPUT ENCODING (MEAN $\pm$ STD OVER FIVE SEEDS).

Model	Enc.	MSE	MAE	R^2
SNN-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
LSTM-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
GRU-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
Transformer-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000

Table XXII

SIENA SCALP EEG (SEIZURE RECORDINGS, 14 SUBJECTS): SNN PRE-PROCESSING MODE / V_{th} / α SELECTED ON EACH FOLD'S VALIDATION BLOCK (PARENTHESED FRACTION = SEEDS NOT UNANIMOUS).

Fold	Test grp.	Val grp.	Enc.	Mode	V_{th}	α
0	—	—	—	—	—	—

Table XXIII

SIENA SCALP EEG (SEIZURE RECORDINGS, 14 SUBJECTS): LSTM-AE / GRU-AE ARCHITECTURE SELECTED BY NSGA-II ON EACH FOLD'S VALIDATION BLOCK – HIDDEN SIZE H , DEPTH L , AND INPUT ENCODING (MEDIAN / MODE OVER THE THREE RE-SCORING SEEDS; FAMILIES DISAMBIGUATED BY THEIR OWN INDEPENDENT SEARCH, NOT BY A SHARED COLUMN).

Family	Fold	Test spk.	Enc.	H	L	val MSE
—	0	—	—	—	—	—

Table XXIV

SIENA SCALP EEG (SEIZURE RECORDINGS, 14 SUBJECTS): TRANSFORMER-AE ARCHITECTURE SELECTED BY NSGA-II ON EACH FOLD'S VALIDATION BLOCK – MODEL WIDTH d_{model} , HEADS, DEPTH N , FEED-FORWARD WIDTH d_{ff} , AND INPUT ENCODING (MEDIAN / MODE OVER THE THREE RE-SCORING SEEDS).

Fold	Test spk.	Enc.	d_{model}	Heads	N	d_{ff}
0	—	—	—	—	—	—

the ranking transfers to a different signal modality (EEG, at $d = 64$ instead of 16) and to two different subject-pool sizes within that modality.

VI. DISCUSSION

Two things should be read from the recording-level tables (Table VII and following) rather than from raw MSE gaps. First,

Table XXV

SIENA SCALP EEG (SEIZURE RECORDINGS, 14 SUBJECTS):
RECORDING-LEVEL PAIRED ANALYSIS. $\bar{d}_r$ = MEAN PER-RECORDING MSE
DIFFERENCE (MODEL — REFERENCE); NEGATIVE FAVOURS THE MODEL. CI
= 95% BOOTSTRAP OVER RECORDINGS; p_{HOLM} = HOLM-CORRECTED
WILCOXON SIGNED-RANK.

Model	Reference	n_{rec}	$\bar{d}_r$	95% CI	p_{HOLM}
—	—	0	0.0000	[0.0000, 0.0000]	1

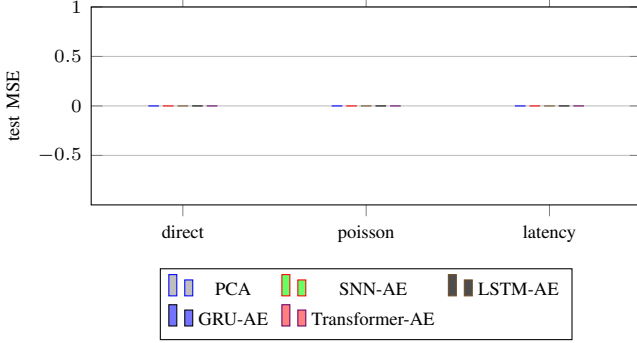


Figure 8. Siena Scalp EEG (seizure recordings, 14 subjects): test MSE by model and input encoding (mean over seeds and folds; mean-frame omitted for scale).

whether each trained family beats the PCA reference at all: under group-disjoint evaluation, a family whose recording-level interval against PCA includes zero has not demonstrated that its nonlinearity buys anything on this benchmark. Second, the ordering among SNN-AE, LSTM-AE, GRU-AE, and Transformer-AE — a difference is only asserted where the Holm-corrected interval excludes zero. The seed-level and group-level checks in the released JSON say whether an ordering that holds in the pooled recording-level analysis also holds run-to-run and group-to-group (speaker-to-speaker or subject-to-subject).

The encoding comparison (the per-dataset per-encoding tables) is where an SNN-specific effect is most likely to appear, because the encoding only reshapes the SNN’s input; the non-spiking baselines consume the same framed window regardless. If latency or Poisson coding helps the SNN relative to direct coding, that is a property of spike-based representation, not of the reconstruction task.

The cost picture is separable from the quality picture. The operation-count column of the summary tables orders the models by arithmetic work under matched software; the self-attention $O(T^2 d_{\text{model}})$ term makes the Transformer’s per-window cost grow faster in sequence length than the recurrent models’, independent of which model reconstructs best. Wall-clock timing on the shared CPU backend is reported alongside, but it reflects this implementation and backend, not silicon-level energy.

Limitations.

- **Few groups.** The four datasets have 6, 60, 109, and 14 groups; the group-level analysis is a robustness check, not population inference, which is why the recording-level

analysis is the primary estimand.

- **Four small datasets.** Two are the same task (spoken digits) and two are EEG (motor imagery and seizure recordings); the ranking may not transfer to other 1-D signal domains. AudioMNIST, EEGMMIDB, and Siena are additionally window-capped per recording (2, 40, and 40) to bound compute. Unlike the earlier MIT-BIH-based design, the EEG group unit is the subject directory itself (Sec. IV), so every recording of a subject is guaranteed to stay in the same split, removing the record-vs-subject over-count concern the earlier design had. The EEG loader reads a single fixed channel (channel 0) of each recording under the assumption that it is an EEG channel and not, e.g., a status channel — verified against real downloaded file headers for both datasets (signal 0 is labelled “EEG Fp1” for Siena and “Fc5.” for EEGMMIDB, both genuine 10–20-system electrode positions).
- **Cost is not energy.** C_{raw} and C_{weighted} count operations under matched software; real energy depends on memory movement, sparsity exploitation, and target hardware. On-device measurement is future work.
- **Correlated windows.** Non-overlapping windows of one recording remain correlated, so window-level statistics are pseudoreplicated and used descriptively only.
- **Bottleneck-matched, not parameter-matched.** SNN-AE, LSTM-AE, GRU-AE, and Transformer-AE each search their own hidden width, depth, and (for the Transformer) attention shape (Sec. III-E); only the latent width d is fixed, per dataset, across all four. Because each family’s search independently picks whatever capacity best trades off validation MSE against inference cost, this is closer to matching *capability under an equal compression ratio* than the fixed-dimension design of the earlier version of this study, but it still does not match trainable parameter count P : dropping the LSTM cell state removes one gate’s worth of parameters per layer, so a GRU-AE winner is smaller than an LSTM-AE winner at the same hidden width; the Transformer-AE’s parameter count follows from its own searched shape, not a size chosen to match the recurrent baselines; and SNN-AE carries its own biophysical parameters (R , C , V_{th}) with no non-spiking analogue. A quality gap can therefore still partly reflect a capacity gap that emerged from the search, not only an inductive-bias difference. A per-family lower bound keeps every searched hidden width and d_{model} at or above d (Sec. III-E), so this capacity difference cannot mean one family’s *effective* bottleneck was narrower than d ; it can still be wider. The summary tables report each model’s real P (Table II and following) precisely so cost and quality can be read together rather than assumed equal.
- **Search budget, not exhaustive search.** Each family’s NSGA-II search evaluates 48 candidates per (dataset, fold, repeat) — a population of 8 for 5 generations, chosen to keep the four families’ search costs comparable in evaluation count rather than wall-clock time, since their per-evaluation costs differ by roughly two orders

of magnitude (Sec. III-E). A larger budget could find a better architecture for any family; the reported comparison is therefore a comparison of what each family reaches under a matched *evaluation* budget, not of each family’s best possible architecture. Likewise, the per-dataset latent width itself (16 for the spoken-digit datasets, 64 for the EEG datasets) follows a brief, non-exhaustive literature scan rather than a systematic sweep on these four datasets: the audio width reflects compression ratios common in prior autoencoder work on short speech; the EEG width follows a single shallow-autoencoder result (hidden size 64) on CHB-MIT, a seizure-EEG dataset in the same task category as Siena but not used in this study (Siena replaced CHB-MIT on 2026-09-23 for dataset-size reasons, after the latent-width literature scan) and extrapolated to EEGMMIDB as well; neither EEG dataset actually used here has a dataset-specific latent-width reference. It is a stated, literature-informed design choice, not a validated optimum.

All results are loaded directly from experiment-generated CSV files; the per-run CSVs are immutable and aggregation never overwrites them.

VII. CONCLUSION

We compared four independently architecture-searched autoencoder families — SNN-AE, LSTM-AE, GRU-AE, and a bottlenecked Transformer-AE — under three spike-encoding front ends, against PCA and mean-frame linear references, for 1-D temporal signal reconstruction on four datasets spanning two signal domains (FSDD, AudioMNIST; the PhysioNet EEG Motor Movement/Imagery Database, the Siena Scalp EEG Database). The evaluation is leakage-controlled: nested six-fold leave-one-group-out cross-validation with no recording or group crossing a split and, for every family, architecture and hyperparameters selected on the inner validation group only via an independent NSGA-II search under a per-dataset latent-width constraint shared by all four families. Differences are reported at the recording level with bootstrap intervals and Holm-corrected tests, with group- and seed-level results as robustness evidence; cost is an analytic per-class operation count, explicitly not an energy measurement. This design directly answers the leakage, missing-baseline, missing-significance, energy-overclaim, and fixed-architecture objections to the earlier version of this study. The harness, profiles, and analysis scripts are released so the protocol can be reused for other 1-D signal families.

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